Grasping Major Statements and their Contradictions Toward Information Credibility Analysis of Web Contents

Daisuke Kawahara[†] Sadao Kurohashi[†‡] Kentaro Inui[†]

[†]National Institute of Information and Communications Technology
3-5 Hikaridai Seika-cho, Soraku-gun, Kyoto 619-0289, Japan
[‡]Graduate School of Informatics, Kyoto University
Yoshida Honmachi, Kyoto 606-8501, Japan

dk@nict.go.jp, kuro@i.kyoto-u.ac.jp, inui@is.naist.jp

Abstract

The World Wide Web contains wide variety of news reports, arguments, opinions, etc. that vary widely in quality. People judge the credibility of information on the Web for decision making in daily life. At present, while the quantity of information on the Web is explosively increasing, it is necessary to develop a system that supports such judgments. We have been developing an information credibility analysis system, WISDOM that considers the viewpoints of information contents, information senders, and information appearances. In this paper, as a viewpoint of information contents, we propose a method for providing a bird's eye view of major statements on a given topic and their contradictions. We evaluate the obtained statements in our experiments, and confirm the effectiveness of our approach. Furthermore, we discuss our future objectives.

1. Introduction

Nowadays, the quantity of information on the Web is explosively increasing, and there is a wide variety of news reports, arguments, opinions, etc. on various topics on the Web. From the viewpoint of the quality and credibility of information, information useful for daily life is mixed with false information and rumors.

People use the Web not only to obtain information on what they want to know but also to collect information for decision making. They generally use conventional search engines such as Yahoo! and Google, but it is difficult to judge the credibility of information from the obtained search results. This is because these search engines rank Web pages in terms of a single measure that is not always related to the credibility of the Web pages. To judge the credibility, users must examine the Web pages individually.

For example, when a user searches for information on "agaricus", which is arguably said to be a health food, using a conventional search engine, commercial pages that advertize its effect on health are highly ranked, and the other pages remain in low ranks. From this search result, it is difficult to judge whether "agaricus" is really good for health.

In order to obtain credible information, it is necessary to develop a new system that automatically processes information on a given topic and provides a bird's eye view and multifaceted views on the topic. For example, these views are a summary page that tabulates reports and opinions on the topic, people and organizations that wrote the pages, whether contact addresses and information sources are specified, and so forth.

We have been developing an information credibility analysis system, WISDOM (Web Information Sensibly and Discreetly Ordered and Marshaled) [3]. WISDOM enables users to browse a large volume of information from a bird's eye view and multifaceted views while changing a search condition on the basis of each factor of information credibility. By using this system, users can obtain credible information on a topic of interest more efficiently.

This paper proposes a method for providing a bird's eye view of major statements on a given topic and their contradictions. This method targets Web pages on the topic, and extracts linguistic expressions occurring with a high frequency as major statements; this method also extracts contradictions to the major statements. This resulting view is a summary that enables users to grasp what facts and opinions are found on the Web. In this summary, not only major statements but also their contradictions are presented even if the number of the contradictions is small. That is to say, this system can extract statements that are related to a given topic including minor ones, and provides users with a bird's

eye view of the topic.

For the topic, *yutori kyouiku* (cram-free education), for example, *gakuryoku-ga teika-suru* (scholastic ability deteriorates) is a major statement and *gakuryoku-ga koujou-suru* (scholastic ability ameliorates) is its contradiction.

2. Grasping major statements and their contradictions

For an analysis of information contents in WISDOM, we propose a method for extracting and presenting major statements on a given topic and their contradictions.

Major statements are defined as linguistic expressions occurring with a high frequency in the set of Web pages on a given topic. Contradictions are the statements that contradict the major statements. For *yutori kyouiku* (cram-free education), for example, *gakuryoku-ga teika-suru* (scholastic ability deteriorates) is a major statement and *gakuryoku-ga koujou-suru* (scholastic ability ameliorates) is its contradiction.

Figure 1 shows a screenshot displaying major statements and their contradictions for *yutori kyouiku* (cram-free education). This display enables users to grasp what statements, such as facts and opinions, are found on the Web. In this summary display, not only major statements but also their contradictions are presented even if the number of the contradictions is small. Therefore, this system can extract statements that are related to a given topic including minor ones, and provides users with a bird's eye view of the topic.

As a unit of statements, we use not a word but a predicate-argument structure (also known as a case structure and logical form). A predicate-argument structure represents an event of "who does what". Processes such as clustering, summarization, comparison with other knowledge and logical consistency verification, which are required for information credibility analysis, are accurately performed on the basis of predicate-argument structures.

Our method consists of the following three steps:

1. extracting predicate-argument structures,
2. merging predicate-argument structures, and
3. detecting major and contradicting predicate-argument structures.

The above processes target Web pages on a given topic. These Web pages are obtained from the open search engine infrastructure, TSUBAKI [6], by inputting the topic as a query. TSUBAKI is a search engine that exploits advanced NLP technologies, which are used also in this study. We retrieve 1,000 Web pages for a topic.

We describe the details of the above three processes in the following sections.

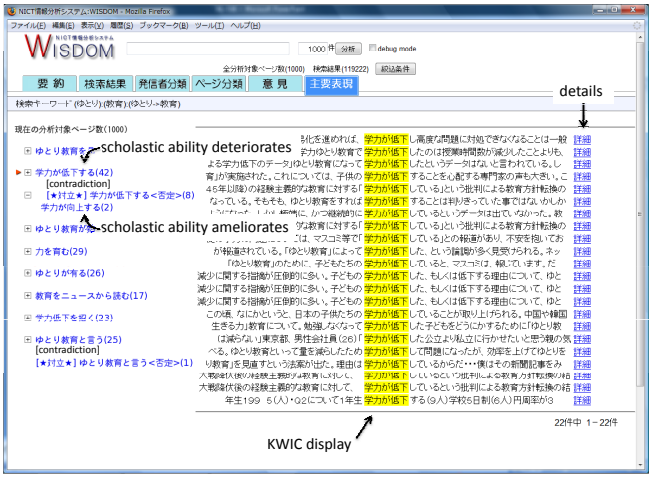

Figure 1. Screenshot of major-contradiction display in WISDOM.

2.1. Extracting predicate-argument structures

A predicate-argument structure consists of a predicate and one or more arguments that have a dependency relation to the predicate.

We extract predicate-argument structures by applying the following procedure to the Web pages that are retrieved from TSUBAKI for a given topic.

1. Extract important sentences from each Web page. Important sentences are defined as the sentences neighboring the topic word(s).

2. Apply a morphological analyzer, JUMAN[1], and a syntactic and case structure analyzer, KNP[2], to the important sentences, and extract predicate-argument structures.

3. Filter functional and meaningless predicate-argument structures, which are not relevant to the topic. These structures can be extracted because we use just parts-of-speech extraction patterns. On the basis of a probability ratio between the entire Web and the target Web pages for a topic, we discard the predicate-argument structures that are not typical of the topic. The following examples are discarded predicate-argument structures: *koto-ni naru* (become ...) and *baai-ga aru* (there is a case ...).

[1] http://nlp.kuee.kyoto-u.ac.jp/nl-resource/juman.html
[2] http://nlp.kuee.kyoto-u.ac.jp/nl-resource/knp.html

2.2. Merging predicate-argument structures

We merge identical and synonymous predicate-argument structures, and count the number of merged predicate-argument structures. Identical predicate-argument structures are simply merged, and furthermore, the following synonymous and subsumed predicate-argument structures are merged.

- Merging synonymous predicate-argument structures

  Based on a synonym lexicon of predicates, the synonymous predicate-argument structures are merged. We use a synonym lexicon that is derived from a Japanese dictionary [5]. This lexicon comprises approximately 6,000 entries. For example, *gakuryoku-ga teika-suru* (scholastic ability deteriorates) and *gakuryoku-ga sagaru* (scholastic ability lowers) are merged, because these predicates are synonyms.

- Merging subsumed predicate-argument structures

  Predicate-argument structures that can be subsumed to other structures are merged. For example, the predicate-argument structure *yutori-de gakuryoku-ga teika-suru* (scholastic ability deteriorates due to relaxation) is subsumed and merged into more general one, *gakuryoku-ga teika-suru* (scholastic ability deteriorates).

  In general, this kind of merging possibly causes the mixing of predicate-argument structures that have different meanings. However, since this task handles only the Web pages about a given topic and the meanings and contexts of predicate-argument structures are narrowed, the proposed merging is assumed not to have a big problem.

2.3. Detecting major and contradicting predicate-argument structures

We extract predicate-argument structures occurring with a high frequency as major ones (i.e., major statements) and predicate-argument structures that contradict the major ones as contradictions.

Contradictions are obtained by searching all the predicate-argument structures that satisfy one of the following conditions.

- Reverse of a negation flag

  If the predicate of a major predicate-argument structure has a negation flag, a contradiction does not have a negation flag. If not, a contradiction has a negation flag. For example, *gakuryoku-ga teika-suru*⟨NEG⟩ (scholastic ability does not deteriorate) is extracted as a contradiction of *gakuryoku-ga teika-suru* (scholastic ability deteriorates). Note that ⟨NEG⟩ represents a negation flag.

- Replacement of a predicate with its antonym

  The predicate of a contradiction is an antonym of that of a major predicate-argument structure. To judge antonym relations, we use an antonym lexicon extracted from a Japanese dictionary for human [5]. This lexicon consists of approximately 2,000 entries. For example, *gakuryoku-ga koujou-suru* (scholastic ability ameliorates) is extracted as a contradiction of *gakuryoku-ga teika-suru* (scholastic ability deteriorates).

3. Experiments

3.1. Experimental settings and results

We evaluated the acquired predicate-argument structures for 25 controversial topics such as *reshikku syujutsu* (LASIK operation), *mainasu ion* (minus ion)[3], *gousei senzai* (synthetic detergent), *yutori kyouiku* (cram-free education), and *NHK jushinryou* (NHK subscription fee).

For each topic, we extracted at most eight major predicate-argument structures in order of frequency. Contradictions are extracted if they exist for each major predicate-argument structure. As a result, 194 major predicate-argument structures and 58 contradictions were obtained. This means that approximately 30% (58/194) of major predicate-argument structures have their contradictions. Table 1 lists some obtained pairs of a major predicate-argument structure and its contradiction.

To evaluate the precision of the obtained predicate-argument structures, we classified them to the following three classes on the basis of the relevance to the given topic.

A: relevant

B: relevant, but should be merged into other predicate-argument structures

C: not relevant

We regard the predicate-argument structures that satisfy all the following conditions as "relevant".

- The target predicate-argument structure is related to the topic.

- The target predicate-argument structure is not a functional/meaningless expression.

[3] "Minus ion" is defined as negative air ions, which are controversially said to be good for health.

Table 1. Examples of obtained major predicate-argument structures and their contradictions.

Topic: *reshikku syujutsu* (LASIK operation)	
syujutsu-wo ukeru (undergo an operation)	
↔	*syujutsu-wo ukeru*⟨NEG⟩ (not undergo an operation)
shiryoku-ga kaihuku-suru (recover sight)	
↔	*shiryoku-ga kaihuku-suru*⟨NEG⟩ (not recover sight)
Topic: *gousei senzai* (synthetic detergent)	
gousei senzai-wo tsukau (use synthetic detergent)	
↔	*gousei senzai-wo tsukau*⟨NEG⟩ (not use synthetic detergent)
kankyou-ni warui (bad for environment)	
↔	*kankyou-ni yoi* (good for environment)

- There is an original sentence whose meaning is the same as the target predicate-argument structure. That is, the predicate-argument structure is correctly extracted.

Table 3 shows some examples of obtained predicate-argument structures and their evaluation results. For *yutori kyouiku* (cram-free education), for example, *gakuryoku teika-wo maneku* (cause deterioration of scholastic ability) is relevant to the topic, but it should be merged into *gakuryoku-ga teika-suru* (scholastic ability deteriorates). So, this predicate-argument structure is judged as "B". On the other hand, *yutori kyouiku-to iu* (as cram-free education) has no meaning, and it was judged as "C".

Table 2 lists the results of the evaluation. From this table, we can see that approximately 80% of the obtained predicate-argument structures are relevant ("A" or "B").

3.2. Discussions

For *yutori kyouiku* (cram-free education), *yutori kyouiku-wo minaosu* (reexamine cram-free education), *gakuryoku-ga teika-suru* (scholastic ability deteriorates), etc. are extracted as major predicate-argument structures. Against *gakuryoku-ga teika-suru* (scholastic ability deteriorates), *gakuryoku-ga koujou-suru* (scholastic ability ameliorates) is extracted as its contradiction. Therefore, we can see that the deterioration and amelioration of scholastic ability are related to cram-free education. We can also see that there are many Web pages that state the deterioration of scholastic ability, but that there are a few Web pages that state the amelioration of scholastic ability, which it is very difficult to find from the search results of conventional search engines. Actually, these pages reported the cases that scholastic ability of students improved due to cram-free education.

Table 2. Accuracy of obtained predicate-argument structures.

	major p-a	contradictions
relevant(A, B)	160/194 (82.5%)	46/58 (79.3%)
relevant(A)	118/194 (60.8%)	39/58 (67.2%)
should be merged(B)	42/194 (21.6%)	7/58 (12.1%)
not relevant(C)	34/194 (17.5%)	12/58 (20.7%)

On the other hand, there are approximately 20% of predicate-argument structures that should be merged (evaluated as "B"). Many of them are attributed to the shortage of the entries in our synonym lexicon. For example, *jushinryou-wo shiharau* (defray subscription fee) and *jushinryou-wo harau* (pay subscription fee) were not merged.

The definition sentence of *shiharau* (defray) is *aite-ni okane-wo harau* (pay money to the other party). Since the whole sentence is regarded as the synonymous phrase, *harau* (pay), which is a part of the definition, was not extracted as a synonym. However, *harau* (pay) can be regarded as the synonym of *shiharau* (defray) under the narrowed context of the target Web pages that are retrieved via the topic. We will verify the effectiveness of this solution in future.

The above method relies on the synonym lexicon. If there is no information to merge synonymous predicate-argument structures in the lexicon, we should consider another knowledge sources such as similarities of context and predicative case frames.

Most of the predicate-argument structures that were judged as "C" were functional and meaningless expressions such as *hajime-ni genkyu-sareru* (... first referred) and *ue-de igibukai* (significant on ...).
These examples were not discarded by the filtering of predicate-argument structures. It is necessary to discard them by adjusting the filtering threshold.

For the topic, *mainasu ion* (minus ion), *mainasu ion-wo hassei-saseru*⟨NEG⟩ (not generate minus ion) is extracted as the contradiction to *mainasu ion-wo hassei-saseru* (generate minus ion), but this was an incorrect extract. This contradicting predicate-argument structure was extracted from the following sentence:

(1) *tada mainasu ion-wo hassei-saseru-nodehanaku,*
just minus ion-ACC not generate

purasu ion-wo osaete mainasu ion-wo huyasu
plus ion-ACC restrict minus ion-ACC increase

Table 3. Examples of evaluation results of obtained predicate-argument structures.

Topic: *yutori kyouiku* (cram-free education)	
yutori kyouiku-wo minaosu (reexamine cram-free education)	A
gakuryoku-ga teika-suru (scholastic ability deteriorates)	A
↔ *gakuryoku-ga koujou-suru* (scholastic ability ameliorates)	A
yutori kyouiku-ga hajimaru (cram-free education starts)	A
chikara-wo hagukumu (cultivate ability)	A
yutori-ga aru (have time)	A
kyouiku-wo nyuusu-kara yomu (read education from news)	A
gakuryoku teika-wo maneku (cause deterioration of scholastic ability)	B
yutori kyouiku-to iu (as cram-free education)	C
↔ *yutori kyouiku-to iu*⟨NEG⟩ (not as cram-free education)	C

koto-ga *juuyou-desu*
(that)-NOM important

(It is important not to just generate minus ions, but to restrict plus ions and to increase minus ions.)

This sentence locally negates the statement, *mainasu ion-wo hassei-saseru* (generate minus ion). However, the negation is a part of the idiomatic expression and the meaning of the whole sentence is affirmative. To deal with such phenomenon, we should consider the use of patterns such as *tada ... dewanaku* (not ... just ...).

4. Related work

We extracted major statements from Web documents. This is similar to the research on multi-document summarization from a viewpoint of information extraction from multiple documents. There have been many studies on multi-document summarization. Among them, in particular, the study of Radev et al. [4] is related to ours. They proposed a method for summarizing multiple documents based on the analysis of synonymy and entailment relations between sentences. Their method and most methods of summarization extract and handle sentences, while our method handles predicate-argument structures, which are finer-grained than sentences.

For merging predicate-argument structures, the most important module is the recognition of synonymous and entailed predicate-argument structures. In recent years, evaluational workshops on recognizing textual entailment (RTE) have been held, and this kind of research has been actively studied [1]. The latest workshop in 2007 made an optional task that recognizes contradictions. Harabagiu et al. also focused on recognizing contradictions [2]. They used negative expressions, antonyms and contrast discourse relations to recognize contradictions. Their method just detects relations between sentences, and does not create a summary.

5. Conclusions

This paper has describes a method for showing a bird's eye view of major statements on a given topic and their contradictions. This method is used as a point of view in an information credibility analysis system, WISDOM. We conducted experiments to evaluate the obtained major statements and their contradictions, and confirmed the effectiveness of our approach. By using our proposed method, we can see a summary of the topic, and can find credible information more efficiently.

The proposed method was simple but achieved a satisfactory accuracy. To improve the method to the level of practical use, it is necessary to absorb a variety of synonymous expressions and to handle more complicated contradictions by integrating a sophisticated method of automatic knowledge acquisition.

References

[1] D. Giampiccolo, B. Magnini, I. Dagan, and B. Dolan. The third PASCAL recognizing textual entailment challenge. In *Proceedings of the ACL-PASCAL Workshop on Textual Entailment and Paraphrasing*, pages 1–9, 2007.

[2] S. Harabagiu, A. Hickl, and F. Lacatusu. Negation, contrast and contradiction in text processing. In *Proceedings of the Twenty-First National Conference on Artificial Intelligence (AAAI-06)*, 2006.

[3] S. Kurohashi. Information credibility criteria project. In *Proceedings of the 1st International Symposium on Universal Communication*, pages 49–52, 2007.

[4] D. R. Radev, H. Jing, M. Styš, and D. Tam. Centroid-based summarization of multiple documents. *Information Processing and Management*, 40:919–938, 2004.

[5] T. Shibata, M. Odani, J. Harashima, T. Oonishi, and S. Kurohashi. SYNGRAPH: A flexible matching method based on synonymous expression extraction from an ordinary dictionary and a web corpus. In *Proceedings of the Third International Joint Conference on Natural Language Processing (IJCNLP2008)*, pages 787–792, 2008.

[6] K. Shinzato, T. Shibata, D. Kawahara, C. Hashimoto, and S. Kurohashi. TSUBAKI: An open search engine infrastructure for developing new information access methodology. In *Proceedings of the Third International Joint Conference on Natural Language Processing (IJCNLP2008)*, pages 189–196, 2008.